

COMPARATIVE ANALYSIS

Computer Vision – Quantitative Performance Comparison

1. Object Detection Performance Comparison

Two object detection models are compared using True Positives (TP), False Positives (FP), and False Negatives (FN). Precision measures how many detected objects are actually correct, while recall measures how many actual objects are successfully detected.

Model	True Positives	False Positives	False Negatives
Model A – Viola–Jones	80	20	30
Model B – YOLO	90	15	20

(a) Precision and Recall Calculation

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

Model A Precision = $80 / (80 + 20) = 80 / 100 = 0.80 = 80\%$.

Model A Recall = $80 / (80 + 30) = 80 / 110 = 0.7273 = 72.73\%$.

Model B Precision = $90 / (90 + 15) = 90 / 105 = 0.8571 = 85.71\%$.

Model B Recall = $90 / (90 + 20) = 90 / 110 = 0.8182 = 81.82\%$.

Model	Precision	Recall
Model A – Viola–Jones	80.00%	72.73%
Model B – YOLO	85.71%	81.82%

(b) Comparison and Overall Performance

Model B has higher precision and higher recall than Model A. Its precision is 85.71%, compared with 80.00% for Model A. Its recall is 81.82%, compared with 72.73% for Model A.

For additional comparison, the F1-score can be calculated as the harmonic mean of precision and recall. Model A has an F1-score of approximately 76.19%, while Model B has an F1-score of approximately 83.72%.

Measure	Model A	Model B	Better Model
Precision	80.00%	85.71%	Model B
Recall	72.73%	81.82%	Model B
F1-score	76.19%	83.72%	Model B

Model B is therefore the better overall object detection model. It produces fewer false positives and fewer false negatives while detecting more true objects. Higher precision means fewer incorrect detections, while higher recall means fewer actual objects are missed.

(c) Justification

Model B is preferred because it performs better on both major evaluation measures. Its higher precision indicates greater detection reliability, and its higher recall indicates better coverage of actual objects. This makes Model B more suitable for applications where both correct detection and reduced missed detections are important.

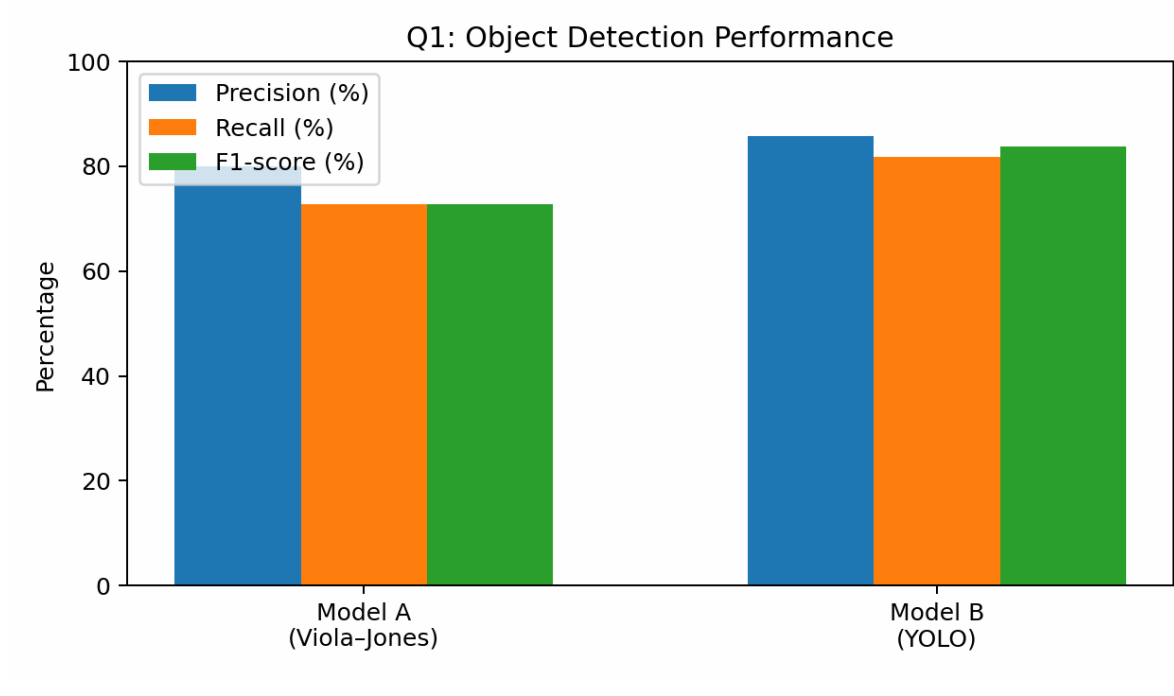


Figure 1: Precision, recall, and F1-score comparison of the two object detection models.

2. Motion Estimation Error Analysis

Motion estimation measures the displacement of an object or image feature between consecutive frames. A lower displacement error indicates that the estimated motion is closer to the true motion.

Frame	Method A Error (pixels)	Method B Error (pixels)
1	2	3
2	3	5
3	2	4

(a) Average Displacement Error

Average Error = Sum of displacement errors / Number of frames

Method A average error = $(2 + 3 + 2) / 3 = 7 / 3 = 2.33$ pixels.

Method B average error = $(3 + 5 + 4) / 3 = 12 / 3 = 4.00$ pixels.

Method	Total Error	Average Error
Method A	7 pixels	2.33 pixels
Method B	12 pixels	4.00 pixels

(b) Comparison

Method A has an average displacement error of 2.33 pixels, whereas Method B has an average error of 4.00 pixels. Since lower error represents more accurate motion estimation, Method A is more accurate.

(c) Justification

Method A has lower error in all three consecutive frames: 2 pixels versus 3 pixels in Frame 1, 3 versus 5 pixels in Frame 2, and 2 versus 4 pixels in Frame 3. Its average error is 1.67 pixels lower than Method B.

The relative reduction in average error is approximately $((4.00 - 2.33) / 4.00) \times 100 = 41.75\%$. Therefore, Method A provides substantially more accurate motion estimates for the given sequence.

Criterion	Method A	Method B
Average error	2.33 pixels	4.00 pixels
Accuracy	Higher	Lower
Consistency	Better	More error
Overall choice	Preferred	Not preferred

Method A should be selected when accurate displacement estimation is the primary requirement. Method B may still be usable in applications where its additional error can be tolerated, but based on the supplied data, Method A is clearly superior.

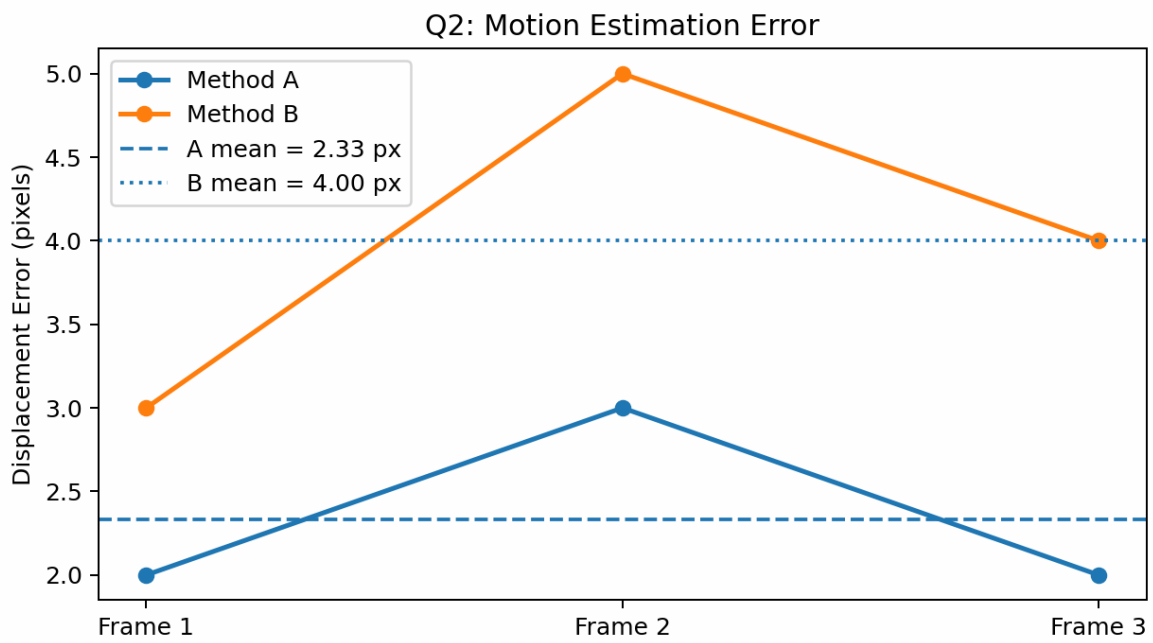


Figure 2: Displacement errors across the three consecutive frames.

3. Depth Estimation Comparison

Depth estimation determines the distance of objects from a camera or stereo system. The accuracy of a depth-estimation method can be evaluated by comparing its estimated depth with the actual depth.

Absolute error is calculated as $|\text{Estimated} - \text{Actual}|$.

Object	Actual Depth (m)	Stereo Vision (m)	SfM (m)
1	2.0	2.0	2.5
2	3.0	3.0	3.8
3	4.0	4.0	5.0

(a) Absolute Error Calculation

Object 1: Stereo error = $|2.0 - 2.0| = 0.0$ m. SfM error = $|2.5 - 2.0| = 0.5$ m.

Object 2: Stereo error = $|3.0 - 3.0| = 0.0$ m. SfM error = $|3.8 - 3.0| = 0.8$ m.

Object 3: Stereo error = $|4.0 - 4.0| = 0.0$ m. SfM error = $|5.0 - 4.0| = 1.0$ m.

Object	Actual (m)	Stereo Error (m)	SfM Error (m)
1	2.0	0.0	0.5
2	3.0	0.0	0.8
3	4.0	0.0	1.0

(b) Average Absolute Error

Stereo Vision average absolute error = $(0.0 + 0.0 + 0.0) / 3 = 0.0$ m.

SfM average absolute error = $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.77$ m approximately.

Method	Average Absolute Error
Stereo Vision	0.00 m
SfM	0.77 m

(c) Comparison and Evaluation

Stereo Vision is more accurate for the supplied data because its estimated depths exactly match the actual depths for all three objects. Therefore, its absolute error is zero for every object.

SfM overestimates the depth of all three objects. Its errors increase from 0.5 m to 0.8 m and then 1.0 m. Its average absolute error is approximately 0.77 m.

The difference between the average errors is 0.77 m, with Stereo Vision providing the smaller error. Therefore, Stereo Vision provides the more accurate depth estimation for this dataset.

Criterion	Stereo Vision	SfM	Better Method
Object 1 error	0.0 m	0.5 m	Stereo
Object 2 error	0.0 m	0.8 m	Stereo
Object 3 error	0.0 m	1.0 m	Stereo
Average absolute error	0.00 m	0.77 m	Stereo

In practical systems, the choice can also depend on scene geometry, camera configuration, baseline, texture, feature availability, and computational requirements. However, based strictly on the supplied measurements, Stereo Vision is the more accurate method.

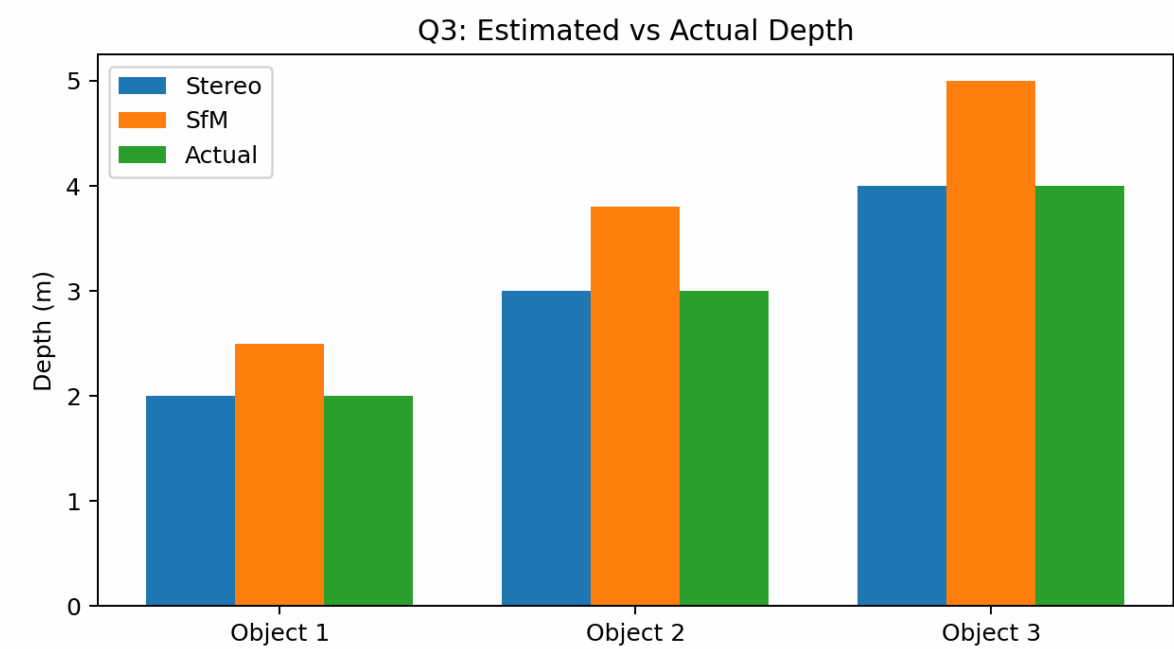


Figure 3: Estimated depth values compared with actual depths.

Overall Conclusion

The quantitative comparisons show that Model B (YOLO) provides better object detection performance because it has higher precision and recall. Method A provides more accurate motion estimation because its average displacement error is lower. Stereo Vision provides the most accurate depth estimates because its average absolute error is zero for the given objects.